

Techniques for reducing GPU usage in WPE

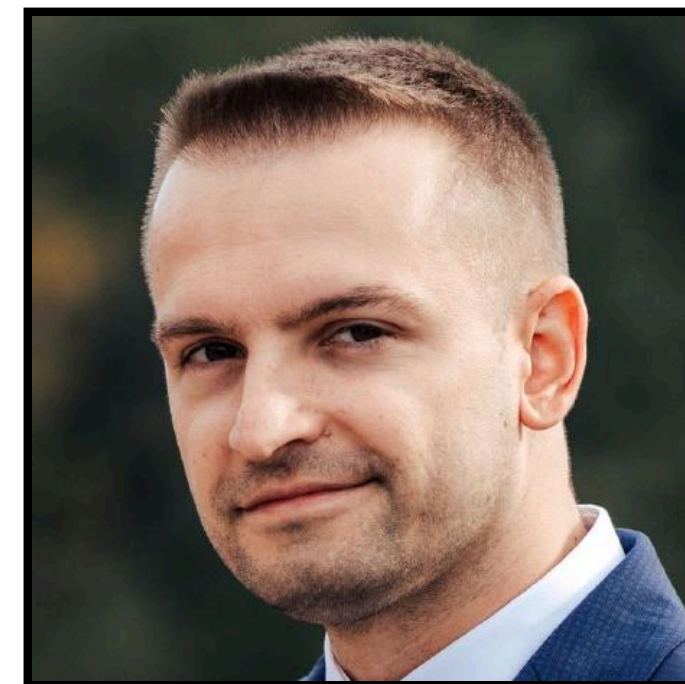
Web Engines Hackfest

June 2026, A Coruña

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About me

- Computer science degree
- Pre-Partner at **Igalia**
- WPE WebKit developer since 2021
 - specialized in Graphics and Multimedia
- Gamedev hobbyist



CPU & GPU in web engines

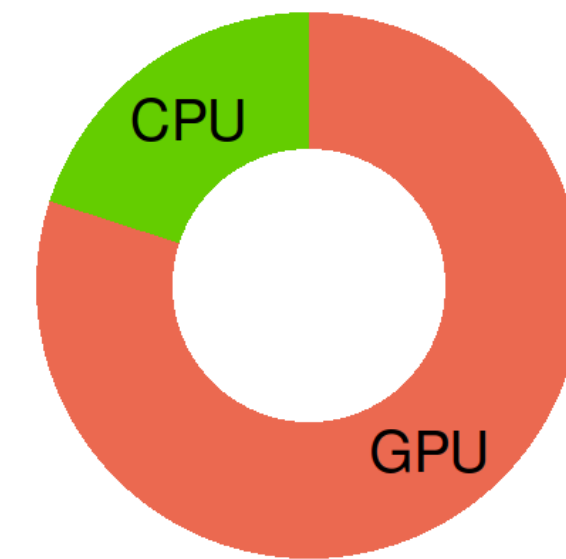
Modern web engines tend to offload as much work to GPU as possible.

- CPU:
 - JavaScript execution
 - DOM processing
 - Style
 - Layout
 - etc.
- GPU:
 - Rendering
 - Compositing



GPU on desktop

- Extra **processing power** with powerful parallelism
- Great for **accelerating** all kinds of rendering
- Dedicated **memory** with bandwidth optimized for batch uploads
- Hardware acceleration for many **primitive operations** such as blending, filtering, etc.

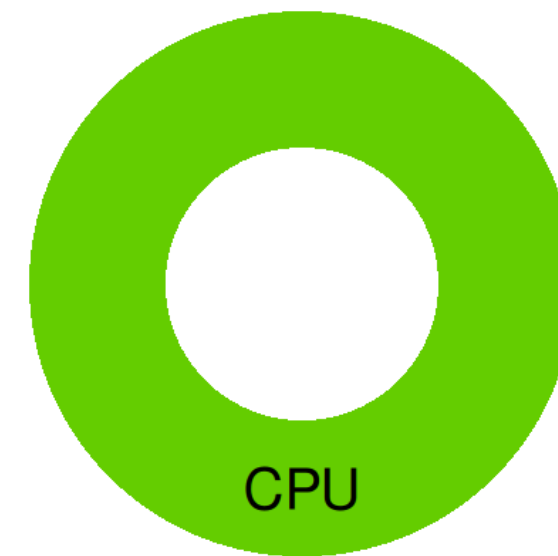
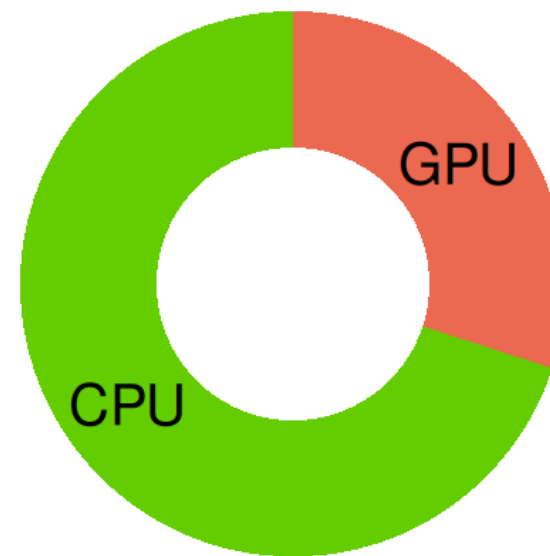
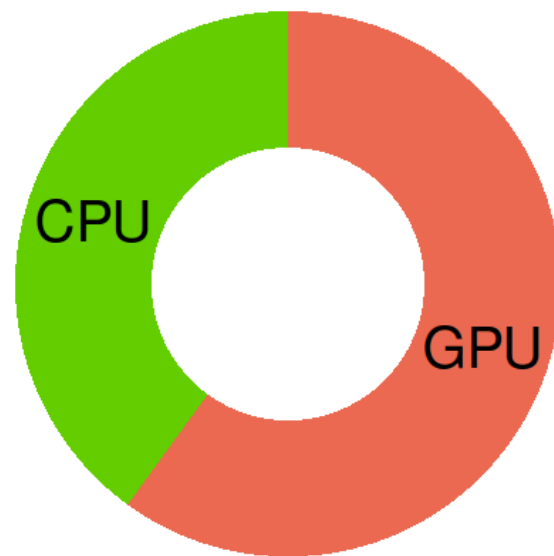


Capabilities



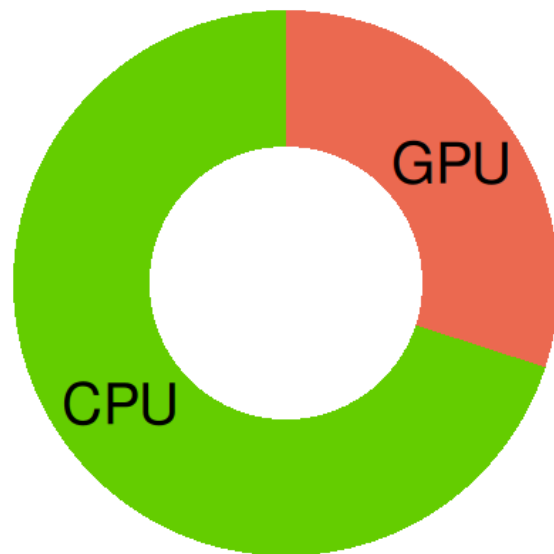
GPU on embedded

- Extra **processing power**
- Good for **accelerating** rendering
- Limited **memory**, shared with CPU
- Hardware acceleration only for some **primitive operations**
- Platform-specific proportions between CPU and GPU capabilities:

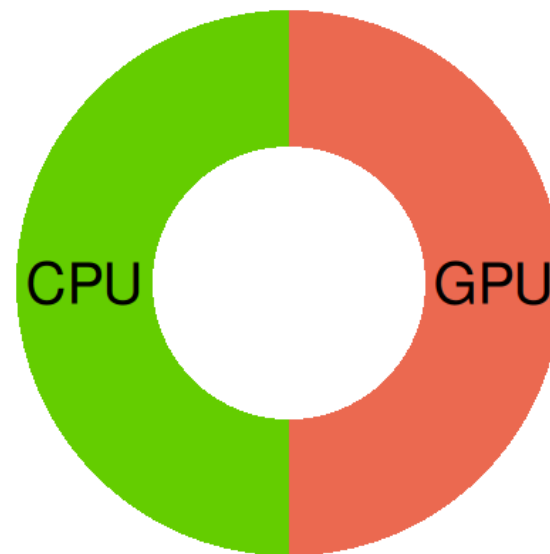


Application requirements

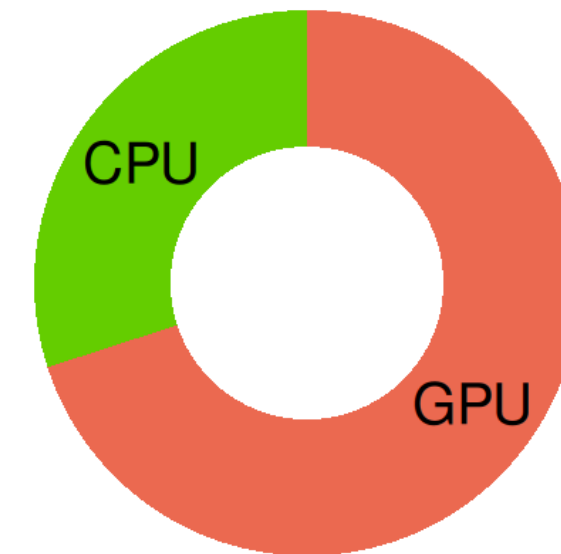
Applications themselves impose different resource usage patterns:



CPU-intensive



Balanced



GPU-intensive



Example 1 - complex & dynamic UI

yahoo/finance

Search for news, tickers or companies

News

Finance

Sports

More

Sign in

My Portfolio

News

Markets

Research

Community

Personal Finance

Videos

Watch Now

Network

US Markets

S&P Futures

Dow Futures

Nasdaq Futures

Russell 2000 Futures

VIX

Gold

Bitcoin USD

Brent Cru

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Trending tickers

BTC-USD

Bitcoin USD

61,622.02

-1,186.16 (-1.89%)

SMCI

Super Micro Com...

40.64

-3.35 (-7.62%)

PAVS

Paranovus Enterta...

1.0200

+0.0795 (+8.45%)

AZI

Autozi Internet Te...

1.8500

+0.7200 (+63.72%)

GC=F

Gold Aug 26

4,214.40

-72.00 (-1.68%)

Rozwijaj klastry kubernetes. Poznaj serwery Scale z...

Serwery OVHcloud

Dowiedz się więcej

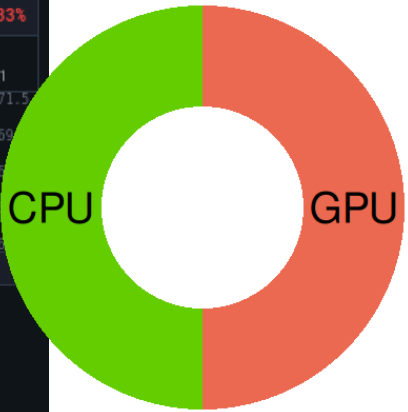
GPU

CPU

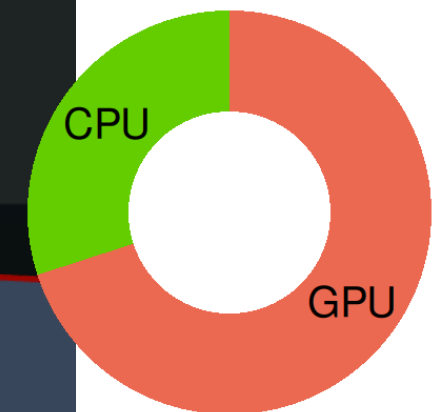
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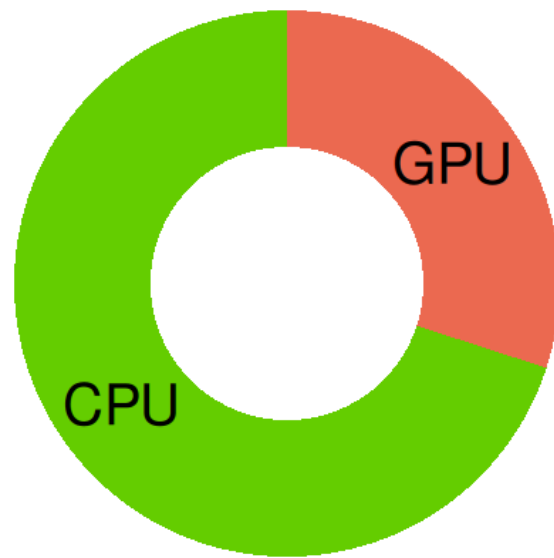
Example 2 - stock dashboard



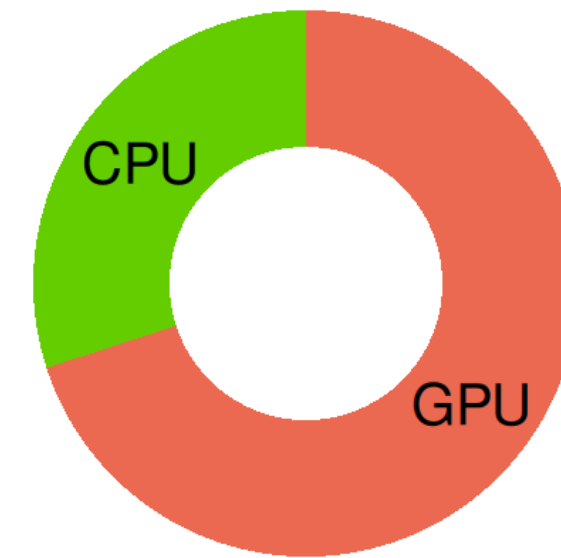
Example 3 - WebGL game



Supply < Demand → Bottleneck



Embedded platform capabilities



Web Application requirements



WPE WebKit overview

- WPE WebKit port is a top web engine for embedded devices
- WPE's default settings target typical embedded use cases:
 - Canvas 2D is mostly accelerated (depends on canvas size)
 - DOM rendering uses GPU
 - Compositor uses GPU (OpenGL)
 - Buffers are shared using DMABuf



Changing the CPU-GPU tradeoff

- WPE offers many ways to change CPU and GPU usage
- Changing CPU and GPU usage is always a **tradeoff**
- There's no silver-bullet set of settings
 - Optimal settings always depend on embedded platform capabilities and web application requirements



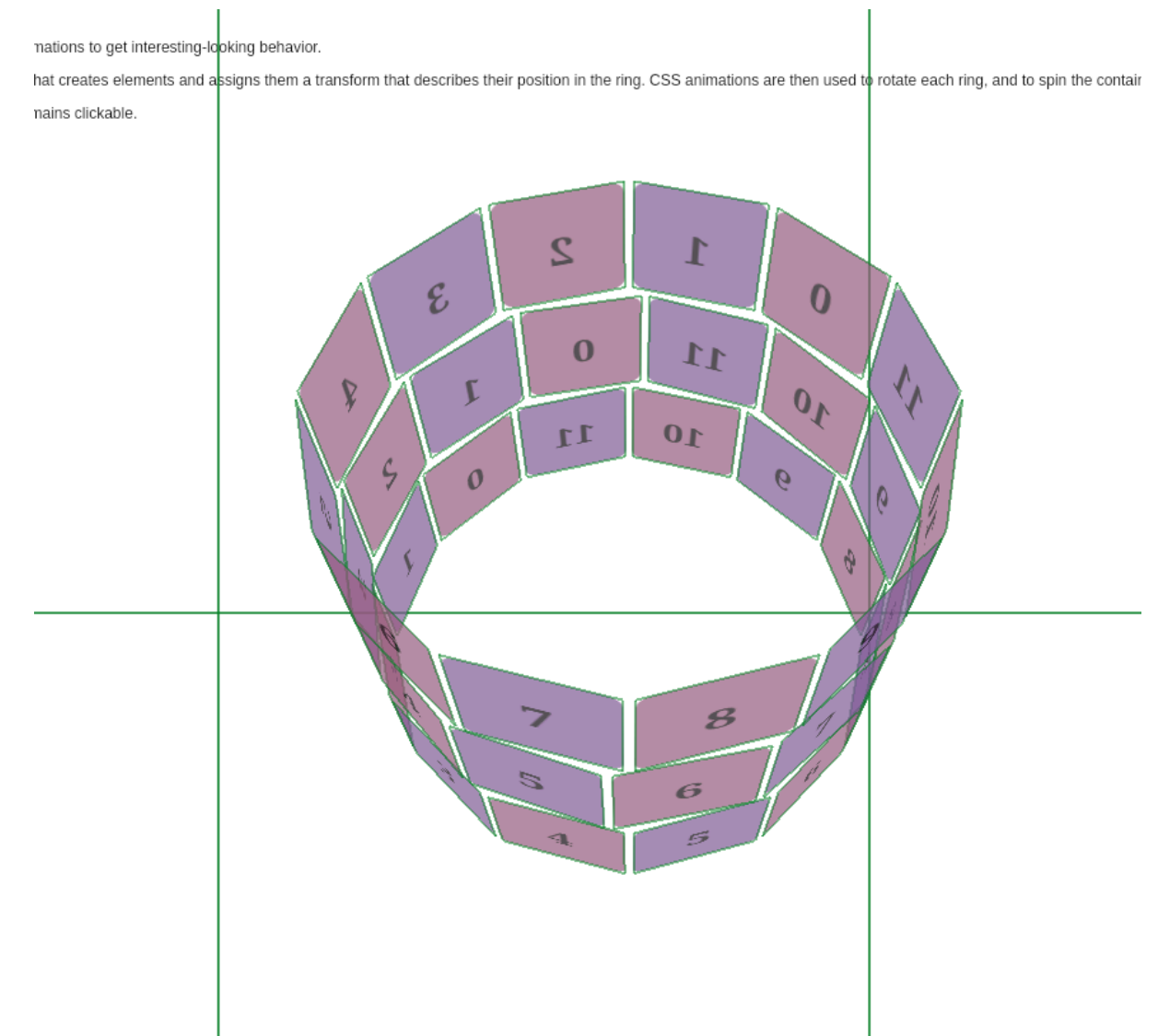
Technique 1: tweaking skia settings

- Utilization: depends on the settings
- Depending on which skia backend is used, some options may change the CPU-GPU tradeoff
- Examples:
 - global options:
 - `GR_AA_TESSELLATOR_MAX_VERB_COUNT`
 - `SK_SUPPORT_LEGACY_CONIC_CHOP`
 - `GrContextOptions`:
 - `fExecutor`
 - `fUseDrawInsteadOfClear`
 - `fInternalMultisampleCount`
 - etc.



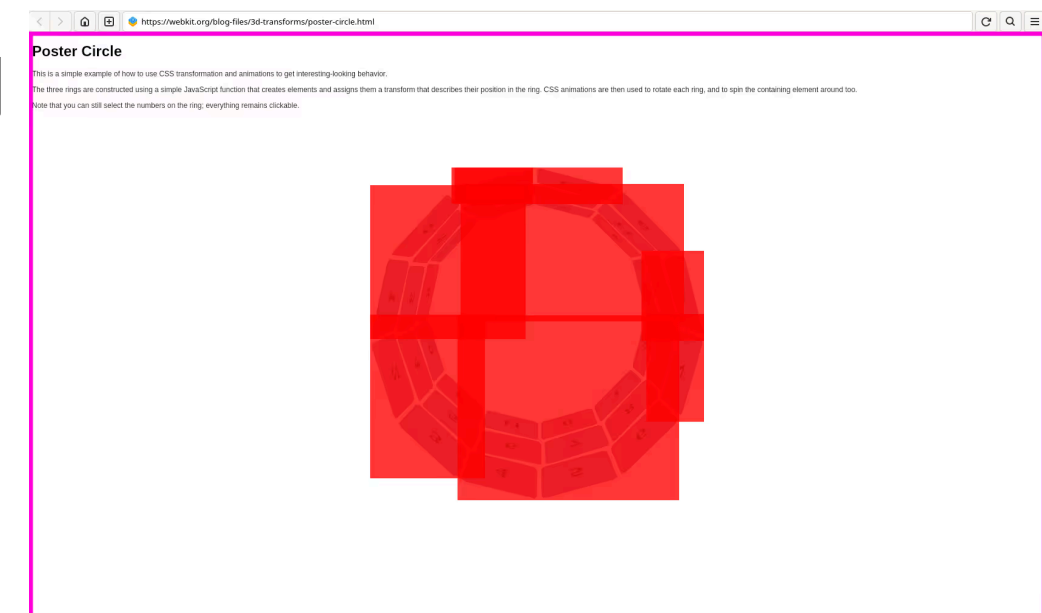
Technique 2: reducing tile sizes

- Utilization: **CPU** , **GPU** 
 - depends on concrete values
- Smaller tile sizes mean:
 - more CPU overhead
 - more GPU draw calls
 - lower GPU memory usage
 - more granular dirty region tracking
 - usually smaller overall repaint area
- Configurable using envvar:
 - `WEBKIT_LAYERS_TILE_SIZE=WIDTHxHEIGHT`



Technique 3: damage propagation ON

- Utilization: **CPU** , **GPU**  
 - depends on damage itself
- Information about changed areas is propagated and:
 - used to reduce composited area in WPE compositor
 - used to hint platform (e.g. wayland compositor)
- Configurable using runtime preferences:
 - `+PropagateDamagingInformation`
 - `+UseDamagingInformationForCompositing`



Technique 4: skia GPU → skia CPU

- Utilization: **CPU**  , **GPU**  
 - depends on DOM rendering intensity
- Skia GPU backend (Ganesh as of 2026) switches to Skia CPU backend
- Effectively all the DOM rendering acceleration is gone
 - Canvas 2D can still be accelerated
 - Compositor still uses GPU (OpenGL)
- Configurable using envvar:
 - `WEBKIT_SKIA_ENABLE_CPU_RENDERING=1`



Technique 5: accelerated 2D canvas OFF

- Utilization: **CPU**  , **GPU**  
 - depends on Canvas 2D usage
- Canvas 2D is neither drawn nor composited in accelerated manner anymore
- Configurable using runtime preference:
 - `-CanvasUsesAcceleratedDrawing`



Technique 6: compositor OFF

- Utilization: **CPU**   , **GPU**  
- Compositing layer tree is disabled
- Compositor is disabled
- All the contents are drawn directly to the final frame
- Configurable using runtime preference:
 - `-AcceleratedCompositing`
- **Experimental in WPE**



Other techniques

- Fine-tuning minor settings
 - `WEBKIT_SKIA_CPU_PAINTING_THREADS`
 - `WEBKIT_SKIA_GPU_PAINTING_THREADS`
 - etc.
- Fine-tuning minor settings in the code
- `LIBGL_ALWAYS_SOFTWARE=1`



The future

- WPE WebKit's compositor is *about to switch from OpenGL to Skia
 - it will open possibility of changing Skia GPU backend from Ganesh to Graphite
 - it will technically make switching to Skia CPU backend doable

*it already did in `main` as of Q2 2026



Thanks!

Questions?



